ATTACHMENTS
Replace Underground Electric
Andrew Johnson National Historic Site

Site Map

Location of Existing Electrical Box

Approx. *Alignment Where Conduits Must Be Chased Into Basement*

Existing Aboveground Electrical Box

Existing Aboveground Electrical Box

Existing Main Disconnect Switch in Basement (shown on left)

Electric As Builts of Homestead Building - Includes Basement & One-Line Diagram

ELECTRICAL LEGEND

	WALL MOUNTED FLUORESCENT STRIP LIGHT FIXTURE, UPPER CASE LETTER INDICATES TYPE. TYPICAL FOR ALL LIGHT FIXTURES.
	1'X4' PENDANT MOUNTED FLUORESCENT INDUSTRIAL LIGHT FIXTURE UNLESS NOTED OTHERWISE. TYPE INDICATED.
	RECESSED DOWN LIGHT FIXTURE, CEILING MOUNTED, OR PENDANT MOUNTED LIGHT FIXTURE, TYPE INDICATED.
	EXTERIOR LIGHT FIXTURE, WALL MOUNTED, TYPE INDICATED.
	PANELBOARD
	EXHAUST FAN OR MOTOR
	DISCONNECT SWITCH.
	CONDUIT ROUTED CONCEALED IN WALL AND/OR ABOVE CEILING
	CONDUIT ROUTED CONCEALED UNDERGROUND OR UNDER FLOOR
	BRANCH CIRCUIT WIRING. A GROUNDING CONDUCTOR, #12 AWG MINIMUM, IS NOT SHOWN BUT SHALL BE PROVIDED. CONDUCTORS SHALL BE 3#12 AWG MINIMUM IN 3/4" CONDUIT MINIMUM UNLESS NOTED OTHERWISE. ARROW TERMINATION INDICATES HOMERUN TO PANEL.
J	WALL/CEILING MOUNTED JUNCTION BOX
	DUPLEX RECEPTACLE FLOOR MOUNTED, UNLESS NOTED OTHERWISE.
	DUPLEX RECEPTACLE, MATCH EXISTING HEIGHT OR IF NEW MOUNT AT 18"AFF, UNLESS NOTED OTHERWISE.
GFCI	DUPLEX RECEPTACLE (GFCI), MATCH EXISTING HEIGHT OR IF NEW MOUNT AT 48" AFF, UNLESS NOTED OTHERWISE.
	DOUBLE DUPLEX RECEPTACLE FLOOR MOUNTED, UNLESS NOTED OTHERWISE.
	DOUBLE DUPLEX RECEPTACLE, MATCH EXISTING HEIGHT OR IF NEW MOUNT AT 18"AFF, UNLESS NOTED OTHERWISE.
S	SINGLE POLE TOGGLE SWITCH MOUNTED AT 48" AFF., UNLESS NOTED OTHERWISE.

ELECTRICAL ABBREVIATIONS:

A	AMPERE/AMPS
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AIC	AMPS INTERRUPTING CAPACITY
C	CONDUIT
CKT	CIRCUIT
EDH	ELECTRIC DUCT HEATER
EF	EXHAUST FAN
E/G	ENGINE/GENERATOR
FA	FIRE ALARM
FU	FUSED
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
HP	HORSEPOWER
HACR	HEATING, AIR CONDITIONING, REFRIGERATION
KVA	KILOVOLT-AMP
LTG	LIGHTING
MDP	MAIN DISTRIBUTION PANEL
MLO	MAIN LUGS ONLY
NEC	NATIONAL ELECTRICAL CODE
P	POLE
PH	PHASE
PP	POWER PANEL
PWR	POWER
RECEPT.	RECEPTACLE
TVSS	TRANSIENT VOLTAGE SUPPRESSION SYSTEM
UV	ULTRA VIOLET
V	VOLTS
W	WATT
WP	WEATHERPROOF
XFMR	TRANSFORMER

ELECTRICAL GENERAL NOTES:

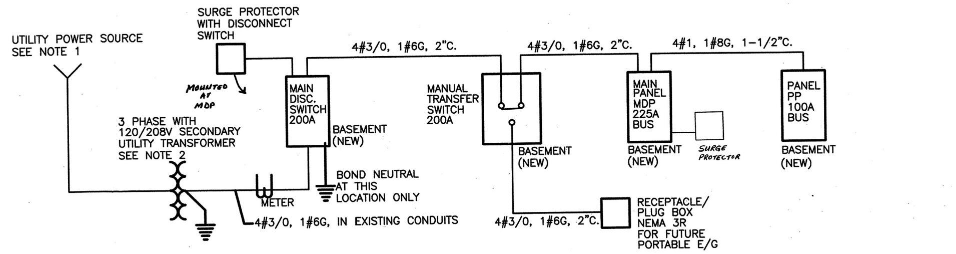
1. REFER TO MECHANICAL DRAWINGS FOR EXACT LOCATIONS AND DIMENSIONS FOR ALL MECHANICAL EQUIPMENTS.
2. ALL ELECTRICAL WORK IN THIS PROJECT SHALL BE IN ACCORDANCE WITH THE 2002 NEC.
3. REFER TO PANEL SCHEDULES, PROVIDE 20A BREAKER AND 2#12, 1#12G, 3/4"C. MINIMUM FOR ALL CIRCUITS UNLESS NOTED OTHERWISE.
4. A (GREEN) EQUIPMEMT GROUNDING CONDUCTOR SHALL BE INCLUDED IN EACH CONDUIT, SIZED IN ACCORDANCE WITH THE 2002 NEC. CONDUIT SHALL NOT BE USED AS GROUNDING CONDUCTOR.
5. MINIMUM TRADE CONDUIT SIZE SHALL BE 3/4". CONDUITS CONTAINING MORE THAN THREE CURRENT CARRYING CONDUCTORS SHALL BE SIZED IN ACCORDANCE WITH THE 2002 NEC.
6. THE DESIGN OF THIS PROJECT IS BASED ON INFORMATION PROVIDED BY THE NATIONAL PARK SERVICE. IF EXISTING CONDITIONS ARE NOT AS DEPICTED OR ANTICIPATED, CONTRACTOR IS TO NOTIFY CONTRACTING OFFICER IMMEDIATELY UPON DISCOVERY AND IS NOT TO PROCEED WITHOUT APPROVAL FROM CONTRACTING OFFICER. CONTRACTOR ACCEPTS FULL LIABILITY FOR THAT PORTION OF THE CONSTRUCTION IF CONTRACTOR PROCEEDS WITHOUT APPROVAL.
7. THE CONTRACTOR IS TO VERIFY ALL EXISTING DIMENSIONS AND CONDITIONS PRIOR TO COMMENCING WITH THE WORK. CONTRACTOR SHOULD STOP WORK IMMEDIATELY IF VARIANCES FROM THE CONSTRUCTION DOCUMENTS, WHETHER REAL OR ANTICIPATED, ARE ENCOUNTERED AND IS TO NOTIFY CONTRACTING OFFICER. FAILURE TO NOTIFY CONTRACTING OFFICER WILL RESULT IN THE CONTRACTOR ACCEPTING FULL LIABILITY FOR THAT PORTION OF THE CONSTRUCTION.

ONE-LINE DIAGRAM NOTES:

1. PRIMARY CONDUCTORS AND CONDUITS SHALL BE PROVIDED AND INSTALLED BY UTILITY COMPANY.
2. CONTRACTOR SHALL COORDINATE WITH UTILITY COMPANY AND NATIONAL PARK SERVICE (NPS) FOR EXACT LOCATION OF TRANSFORMER AND SCHEDULE FOR ENERGIZING NEW ELECTRICAL SERVICE.

LIGHT FIXTURE SCHEDULE

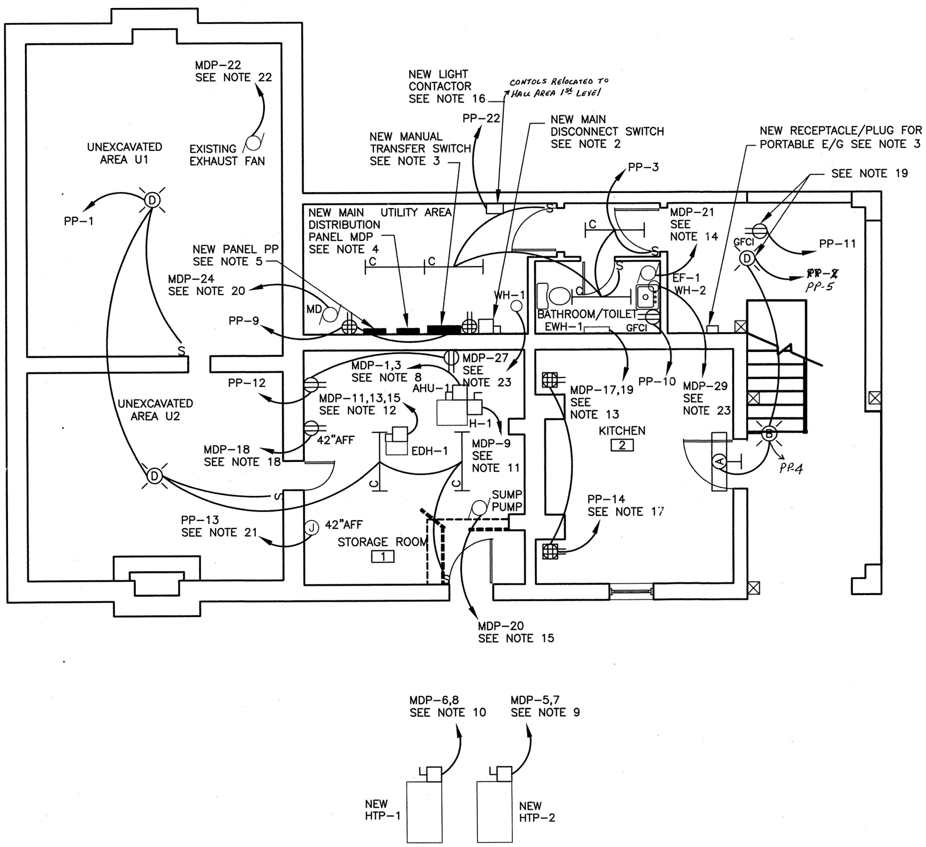
TYPE	CATALOG NUMBER	VOLT	LAMPS	NOTES
A	4' LENGTH FLUORESCENT WALL BRACKET LIGHT FIXTURE, UP-LIGHT COLUMBIA MODEL #COV-4-2-32-EB8-120-EL	120	(2) 32W T8	DIMMABLE WITH UV LAMP FILTERS AND SWITCH
B	EXISTING EXTERIOR LANTERN LIGHT FIXTURE WITH INCANDESCENT LAMP	120	(1) 75W A23	RE-WIRE TO BE REUSED
C	4' LENGTH INDUSTRIAL FLUORESCENT LIGHT FIXTURE, PENDANT MOUNTED AT 9'-0"AFF, DAY BRITE #IA-2-32-120-1/2-EB	120	(2) 32W T8	WITH UV LAMP FILTERS
D	FLUORESCENT SURFACE MOUNTED LIGHT FIXTURE WITH GLASS GUARD COOPER LIGHTING, CROUSE HINDS #VFA122GP	120	(2) 9W	WITH COMPACT FLUORESCENT LAMPS
E	6" RECESSED INCANDESCENT DOWNLIGHT FIXTURE, FOR DAMP LOCATIONS COOPER LIGHTING, PORTFOLIO #PD6-V120-6VWB	120	(1) 150W A19	
F	EXISTING HISTORIC LIGHT FIXTURE WITH INCANDESCENT LAMP	120	(1) 75W A23	RE-WIRE TO BE REUSED



1/E1 **ONE-LINE DIAGRAM**
NOT TO SCALE

DESIGNED: ND	SUB SHEET NO.	TITLE OF SHEET
CADD: ND	E1	ELECTRICAL LEGEND, ABBREVIATIONS & NOTES
TECH. REVIEW: AD		ANDREW JOHNSON NATIONAL HISTORIC SITE
DATE: 2004 APRIL		

DRAWING NO. 365 80,026
PKG. NO.
SHEET 26 OF 40



1 E2 BASEMENT PLAN – ELECTRICAL NEW WORK

NOTES:

1. REFER TO DRAWING E1 FOR LEGEND, GENERAL NOTES AND LIGHTING FIXTURES SCHEDULE.
2. PROVIDE AND INSTALL NEW MAIN DISCONNECT SWITCH, 200 AMPS., 208V, CONNECT TO EXISTING CONDUIT STUB-UP. REFER TO DRAWING E1 FOR ONE-LINE DIAGRAM.
3. PROVIDE AND INSTALL NEW MANUAL TRANSFER SWITCH, 200 AMP., AND EXTERIOR NEMA 3R RECEPTACLE/PLUG OUTLET FOR PORTABLE ENGINE/GENERATOR. REFER TO ONE-LINE DIAGRAM DRAWING E1.
4. PROVIDE AND INSTALL NEW MAIN DISTRIBUTION PANEL MDP, 225A BUS RATING, 120/208V, 3 PHASE, 4 WIRES, 42 CIRCUITS WITH 200A/3P MAIN CIRCUIT BREAKER FED FROM NEW MAIN DISCONNECT SWITCH THROUGH NEW MANUAL TRANSFER SWITCH LOCATED IN BASEMENT. REFER TO DRAWING E1 FOR ONE-LINE DIAGRAM AND DRAWING E6 FOR PANEL SCHEDULE.
5. PROVIDE AND INSTALL NEW PANEL PP, 100A BUS RATING, 120/208V, 3 PHASE, 4 WIRES, 42 CIRCUITS, MAIN LUG ONLY, FED FROM NEW MAIN PANEL MDP. REFER TO ONE-LINE DIAGRAM DRAWING E1 AND DRAWING E6 FOR PANEL SCHEDULE.
6. ALL EXISTING CONDUITS THROUGHOUT THE BUILDING SHALL REMAIN AND BE REUSED AS MUCH AS POSSIBLE UNDER NEW WORK.
7. REFER TO MECHANICAL DRAWINGS FOR EXACT LOCATIONS AND DIMENSIONS.
8. AHU-1: 1HP, 208, 1PH. PROVIDE 20A/2P HACR CIRCUIT BREAKER, 30A/2P DISCONNECT SWITCH AND 2#12, 1#12G, 3/4"C.
9. HTP-2: 208V, 1PH. PROVIDE 25A/2P HACR CIRCUIT BREAKER, 30A/2P DISCONNECT SWITCH AND 2#10, 1#10G, 3/4"C.
10. HTP-1: 208V, 1PH. PROVIDE 40A/2P HACR CIRCUIT BREAKER, 60A/2P DISCONNECT SWITCH AND 2#8, 1#10G, 3/4"C.
11. H-1: 1.4KW, 120V, 1PH. PROVIDE 15A/1P HACR CIRCUIT BREAKER, 30A/1P DISCONNECT SWITCH AND 2#12, 1#12G, 3/4"C.
12. EDH-1: 10KW, 208V, 3PH. PROVIDE 40/3P HACR CIRCUIT BREAKER, 60A/3P DISCONNECT SWITCH AND 3#8, 1#10G, 3/4"C.
13. EWH-1: 2KW, 208V, 1PH. PROVIDE 20A/2P HACR CIRCUIT BREAKER AND 2#12, 1#12G, 3/4"C.
14. EF-1: 0.7A, 120V, 1PH. PROVIDE 20A/1P HACR CIRCUIT BREAKER, MOTOR RATED SWITCH AND 2#12, 1#12G, 3/4"C.
15. SUMP PUMP: 1/2HP, 120V, 1PH. PROVIDE 20A/1P HACR CIRCUIT BREAKER, 2#12, 1#12G, IN EXISTING CONDUITS.
16. LIGHT CONTACTOR TO CONTROL ALL LIGHT FIXTURES IN THE BUILDING. REFER TO DIAGRAM DRAWING E6.
17. UTILIZE EXISTING STUB-UP CONDUITS.
18. AIR COMPRESSOR: 120V, 1PH. PROVIDE 20A/1P HACR CIRCUIT BREAKER, AND 2#12, 1#12G, 3/4"C. COORDINATE EXACT LOCATION AND HEIGHT WITH FIRE PROTECTION DRAWINGS.
19. EXTERIOR LIGHT FIXTURE AND DUPLEX RECEPTACLE MOUNTED IN SAME LOCATION AS EXISTING, ON UPPER JOISTS, REUSE EXISTING CONDUITS.
20. MOTORIZED DAMPER (MD): 120V, 1PH. PROVIDE 20A/1P HACR CIRCUIT BREAKER, MOTOR RATED SWITCH AND 2#12, 1#12G, 3/4"C.
21. FIRE ALARM CONTROL PANEL (FACP): 120V, 1PH. PROVIDE 20A/1P CIRCUIT BREAKER, 2#12, 1#12G., 3/4"C. COORDINATE EXACT LOCATION AND HEIGHT WITH FIRE PROTECTION DRAWINGS.
22. EXISTING EXHAUST FAN: 120V, 1PH. PROVIDE 20A/1P HACR CIRCUIT BREAKER, MOTOR RATED SWITCH AND 2#12, 1#12G, IN EXISTING CONDUIT.
23. WH-1 AND WH-2: 2.4KW, 120V, 1PH. EACH. PROVIDE 30A/1P HACR CIRCUIT BREAKER, 30A/1P DISCONNECT SWITCH AND 2#10, 1#10G, 3/4"C. FOR EACH.

4 0 4 8
SCALE OF FEET

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
ND	E2	BASEMENT PLAN ELECTRICAL NEW WORK	365 80,026
CADD ND			PKG. NO.
TECH. REVIEW: AD		ANDREW JOHNSON NATIONAL HISTORIC SITE	SHEET 27 OF 40
DATE: 2004 APRIL			

Greenville Electric Authority – Standard Underground Electric Service Requirements to a Structure

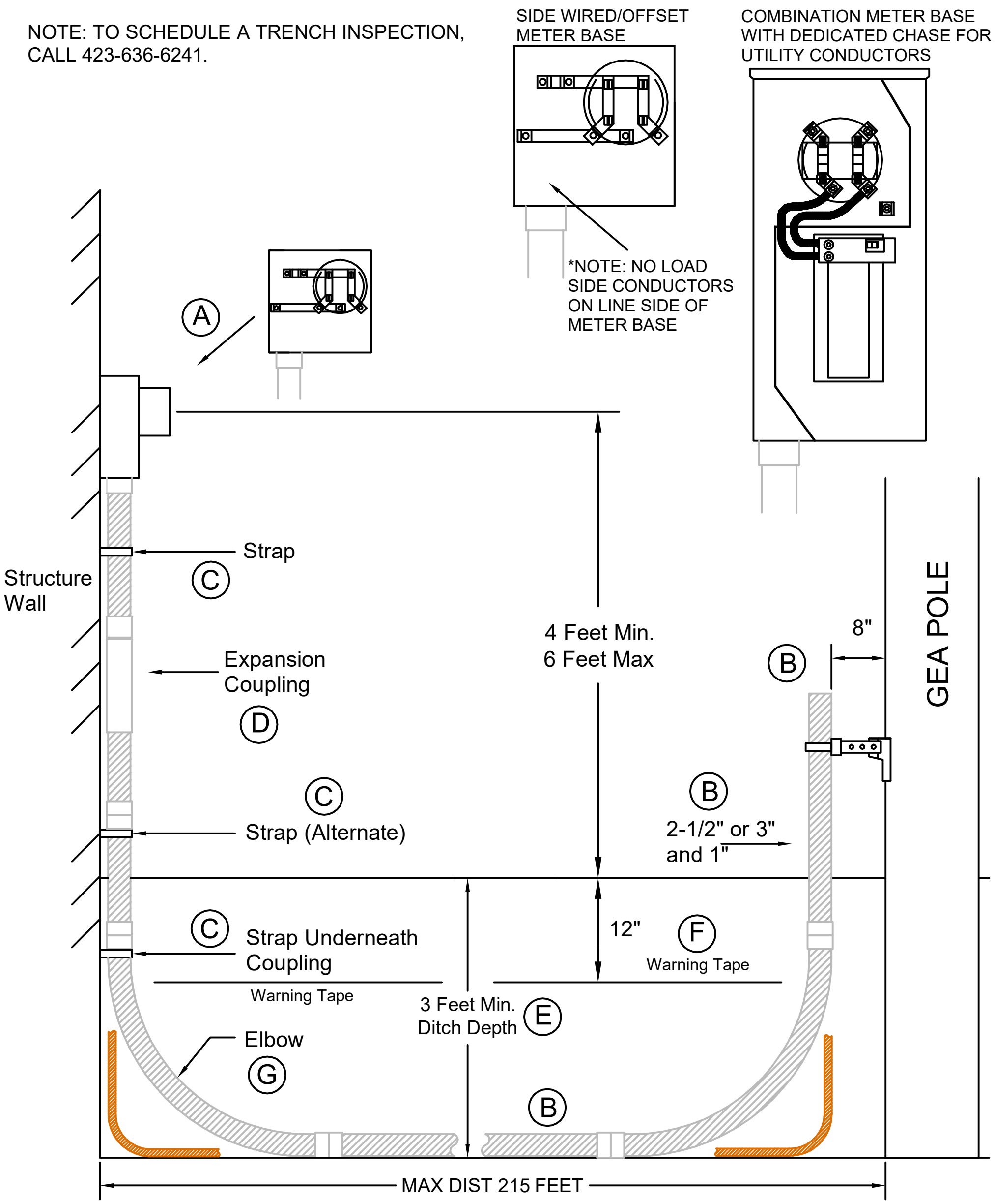
STANDARD UNDERGROUND ELECTRIC SERVICE REQUIREMENTS TO A STRUCTURE

100-200 AMP 120/240 Volt Single Phase

**THE REQUIREMENTS OF THIS SPECIFICATION ARE EFFECTIVE JANUARY 2, 2024 AND SUPERSEDE ALL PREVIOUS PUBLICATIONS

NOTE: BEFORE GEA INSTALLS ANY FACILITIES ON PRIVATE PROPERTY, AN EASEMENT WILL NEED TO BE OBTAINED

NOTE: TO SCHEDULE A TRENCH INSPECTION, CALL 423-636-6241.



*NOTE: 1 INCH COMMUNICATION DUCT NOT SHOWN IN ITS ENTIRETY FOR CLARITY

Greeneville Energy Authority
423-636-6200
www.mygea.net

A. **UG METER BASE:** Customer must contact GEA Engineering Department at 423-636-6241 to have the meter base location determined. GEA may require relocation of the meter base when customer has not complied with this requirement. Meter base must be mounted a minimum of 4 feet and a maximum of 6 feet above finished grade. Meter base must be grounded per National Electric Code requirements.

NOTE: ONLY SIDE WIRED/OFFSET METER BASES OR METER BASE COMBINATIONS WITH A DEDICATED CHASE FOR UTILITY SIDE CONDUCTORS WILL BE ACCEPTED FOR GEA TO FURNISH AND INSTALL THE UNDERGROUND SERVICE CONDUCTORS TO THE METER BASE.

B. **CONDUIT:** All underground services shall be installed in conduit regardless of soil conditions. ***NO LB'S, LL'S OR LR'S ARE PERMITTED ON THE LINE SIDE OF METER BASE.***
- Customer to furnish and install all conduit from meter base to GEA pole per drawing. *Maximum underground service length from GEA pole is 215 feet.* All joints to be glued.
- 2-1/2 inch or 3 inch UL listed Schedule 40 PVC conduit is required for the electrical service. No reducers or mixing of conduit sizes are permitted. Customer must also furnish and install a 1 inch UL listed Schedule 40 PVC conduit for future communications/meter reading. The 1 inch conduit is to be installed in close proximity and preferably in contact with the electrical service conduit. ***NOTE: THE 1 INCH CONDUIT CANNOT BE USED BY ANY OTHER ENTITY DUE TO ITS INSTALLED LOCATION. THE CONDUIT WILL NOT MEET NESC RULE 320B2c FOR MINIMUM SEPARATION REQUIREMENTS.***
- **GEA WILL NOT ACCEPT HEATED CONDUIT REGARDLESS OF THE METHOD USED. UNDER NO CIRCUMSTANCES WILL GEA ACCEPT CONDUIT OFFSETS TO AVOID NOTCHING/MODIFYING OF FOOTER AT STRUCTURE.**
- All Conduits are to be installed above grade and must not exceed 2 inches away from structure wall. Conduit at GEA pole to be installed in contact with GEA stand-off bracket. If GEA stand-off bracket is not present during installation, install conduit 8 inches away from surface of pole. Seal the end of all conduits that do not enter an enclosure.

C. **COUPLING SUPPORT STRAP/STRAPS:** Install strap directly below coupling ensuring contact with coupling. This prevents conduit from settling during backfill. If there is nothing below grade to anchor the coupling support strap, cut the conduit one foot above grade and install coupling support strap. Straps are required every 24 inches for exposed conduit above grade.

D. **EXPANSION COUPLING:** An expansion coupling is required for all underground services.

E. **TRENCH 3 FEET DEEP MINIMUM:** Customer to open and close service trench. Depth of trench to be a minimum of 3 feet. If rock is encountered and minimum depth cannot be achieved, conduit can be encased with 3 inches of concrete. Water lines cannot be installed in the trench with electrical lines unless installed by "shelf method" as specified by GEA drawing JT1-722. If "shelf method" was not utilized, a minimum of 5 feet of separation is required. Water lines can cross GEA lines as long as a minimum of 12 inches of separation is maintained. Communication lines can be installed in the same trench as electric as long as a minimum of 12 inches of separation is maintained. Refer to "GEA TRENCH REQUIREMENTS" located at www.mygea.net or obtain a copy from GEA Engineering Department. Trench must be backfilled before GEA can energize service.

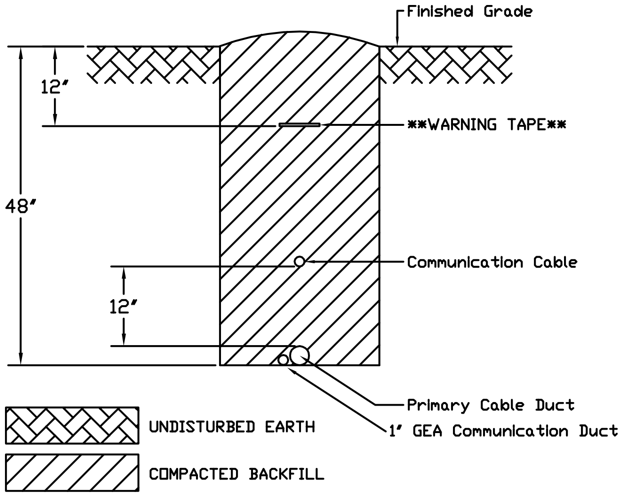
F. **WARNING TAPE AND TRENCH INSPECTION:** Warning Tape must be installed and secured to all vertical elbows. The warning tape can be in the trench or on the ground beside the trench at time of inspection. GEA will not install facilities unless warning tape is present. To schedule a trench inspection, contact GEA Engineering Department at 423-636-6241.

G. **ELBOW:** Must use at least 24 inch radius, 90-degree elbows at GEA pole and at meter base on structure. 36 inch radius 90-degree elbows must be used for horizontal turns in trench. Maximum of 3 elbows (one at pole, one at structure and one in trench) can be used. For the communication conduit, use 5.75 inch radius (standard) elbows.

NOTE: ALL REQUIREMENTS LISTED ABOVE MUST BE MET FOR GEA TO FURNISH AND INSTALL UNDERGROUND CONDUCTORS TO METER BASE. IF THE REQUIREMENTS CANNOT BE MET, REFER TO THE GEA SPECIFICATION "NON-STANDARD UNDERGROUND ELECTRIC SERVICE REQUIREMENTS TO A STRUCTURE".

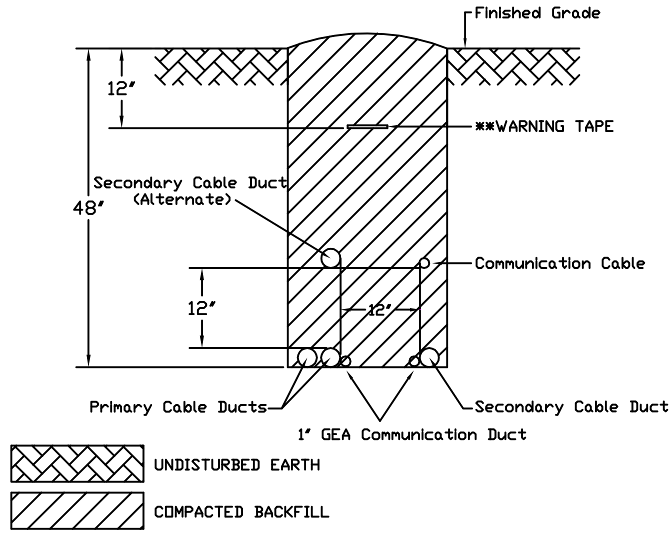
Greenville Electric Authority – Trench Requirements

PRIMARY DUCT WITH COMMUNICATION



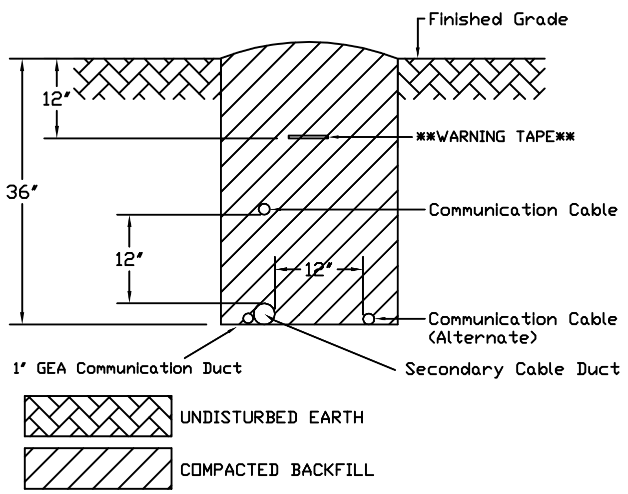
*NOTE: PRIMARY CABLE CONDUITS SHALL BE INSTALLED 48" BELOW FINISHED GRADE.
*NOTE: 12" (MEASURED VERTICALLY) OF WELL TAMPED EARTH SHALL SEPARATE GEA PRIMARY CONDUITS AND COMMUNICATION CABLES.

PRIMARY/SECONDARY DUCT WITH COMMUNICATION



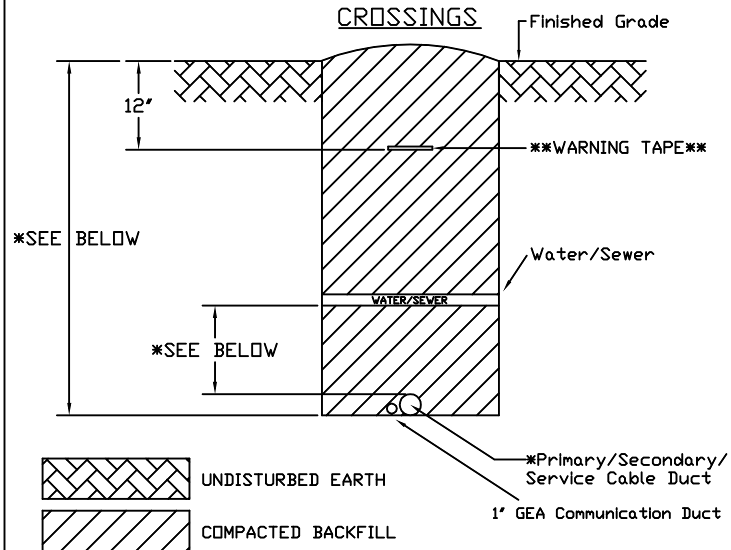
*NOTE: PRIMARY CABLE CONDUITS SHALL BE INSTALLED 48" BELOW FINISHED GRADE.
*NOTE: 12" (MEASURED VERTICALLY) OF WELL TAMPED EARTH SHALL SEPARATE GEA PRIMARY CONDUITS AND COMMUNICATION CABLES. 12" (MEASURED HORIZONTALITY) OF WELL TAMPED EARTH SHALL SEPARATE GEA SECONDARY/ /SERVICE CONDUITS FROM COMMUNICATION CABLES.

SERVICE/SECONDARY DUCT WITH COMMUNICATION



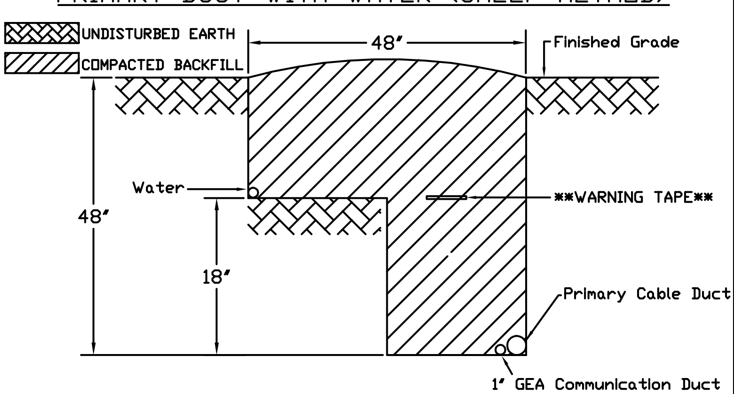
*NOTE: SECONDARY / SERVICE DUCTS SHALL BE INSTALLED 36" BELOW FINISHED GRADE.
*NOTE: 12" (MEASURED VERTICALLY OR HORIZONTALITY) OF WELL TAMPED EARTH SHALL SEPARATE GEA SECONDARY CONDUITS AND COMMUNICATION CABLES.

CROSSINGS



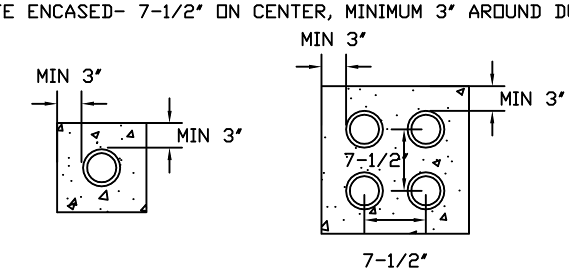
*NOTE: MINIMUM PRIMARY CONDUIT DEPTH IS 48". MINIMUM SECONDARY/ SERVICE CONDUIT DEPTH IS 36".
*NOTE: A MINIMUM OF 12" (MEASURED VERTICALLY) OF WELL TAMPED EARTH SHALL SEPARATE GEA CONDUITS AND OTHER UTILITIES SUCH AS WATER AND SEWER.
*NOTE: WHEN SUCH A CROSSING IS NECESSARY, GEA CONDUIT SHOULD BE AT THE LOWER LEVEL IN TRENCH.

PRIMARY DUCT WITH WATER (SHELF METHOD)



ENCASEMENT

INSTANCES WHEN MINIMUM DEPTHS CANNOT BE ACHIEVED, DUCT TO BE CONCRETE ENCASED- 7-1/2" ON CENTER, MINIMUM 3" AROUND DUCT



WARNING TAPE & SCHEDULING TRENCH INSPECTION

*WARNING TAPE MUST BE INSTALLED AND SECURED TO ALL VERTICAL ELBOWS. THE WARNING TAPE CAN BE IN TRENCH OR ON THE GROUND BESIDE THE TRENCH AT TIME OF INSPECTION. GEA WILL NOT INSTALL FACILITIES UNLESS WARNING TAPE IS PRESENT.

*TO SCHEDULE A TRENCH INSPECTION, CONTACT GEA ENGINEERING DEPARTMENT AT 423-636-6241.

GEA

DWN SWB
DATE 10-13-97
CHK
DATE
APP
DATE

NO	DATE	REVISION
1	05-22	1" COMM DUCT
2	07-22	WATER & GEA

SCALE NONE
SHT# 1 OF 1

SUBJECT DETAIL:
TYPICAL JOINT TRENCH

DWG #
JT1-722

ENGINEERING DEPT.